

Information Communication Technology

Course: ATD — Accounting Technicians Diploma

Level: Level I

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Syllabus version: Based on the standard KASNEB ATD Level I Information Communication Technology syllabus topics — computer fundamentals, application software, and data/information security. Authored from general ICT knowledge and standard syllabus topic coverage, not copied from any existing notes provider or study guide. Cross-check terminology and emphasis against the current KASNEB syllabus and examiner's reports before the exam.

Original work notice. Every explanation, worked example, and question in this document was written from scratch for this paper. No text has been copied or paraphrased from any KASNEB past paper, textbook, or third-party notes/study guide. The scenarios and practice questions are original.

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1. Computer fundamentals — hardware

Hardware refers to the physical, tangible components of a computer system:

- **Input devices** — keyboard, mouse, scanner — used to enter data into the computer.
 - **Output devices** — monitor, printer, speakers — used to present processed information to the user.
 - **The central processing unit (CPU)** — often called the "brain" of the computer, it carries out instructions and performs calculations.
 - **Storage devices** — hard disk drives, solid-state drives, flash drives — used to store data and programs, distinguished from the computer's temporary working memory (RAM).
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2. Computer fundamentals — software

Software refers to the intangible instructions/programs that direct hardware to perform tasks:

- **System software** — the operating system (e.g. Windows) that manages hardware resources and provides the platform on which other software runs.
 - **Application software** — programs designed for specific user tasks (word processors, spreadsheets, accounting packages, presentation software).
 - **Utility software** — programs supporting system maintenance (antivirus, disk clean-up, backup tools).
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3. Data versus information

Data is raw, unprocessed facts and figures (e.g. a list of individual sales transactions) with no meaning on its own until organised. **Information** is data that has been processed, organised and structured so it becomes meaningful and useful for decision-making (e.g. a monthly sales summary report). Good information should generally be: accurate, timely, relevant, complete, and presented in a format the user can readily understand and use.

4. Common application software used in accounting/business

- **Word processors** (e.g. for letters, reports) — creating and formatting text documents.
- **Spreadsheets** (e.g. for budgets, simple ledgers, calculations) — organising data into rows and columns, with built-in formulas and functions.
- **Accounting/ERP packages** — specialised software for recording transactions, generating financial statements, and managing inventory/payroll.
- **Presentation software** — creating slide-based presentations for meetings and reports.

- **Database software** — storing and retrieving structured data efficiently, useful for managing larger volumes of customer or transaction records.
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5. Spreadsheets for basic accounting tasks

A spreadsheet organises data into a grid of **rows** and **columns**, with each intersection called a **cell**. Common uses in accounting-support work include: simple budgets, basic ledger summaries, loan/interest calculations, and straightforward "what-if" analysis (changing one input to see the effect on a result). Common built-in functions include SUM (adding a range of values), AVERAGE (computing the mean of a range), and simple formulas combining cell references with basic arithmetic operators.

6. Computer networks, basics

A **network** connects multiple computers to share data, software and hardware resources (e.g. a shared printer). Common types:

- **LAN (Local Area Network)** — covers a small geographic area, e.g. a single office building.
- **WAN (Wide Area Network)** — covers a large geographic area, potentially connecting offices in different cities or countries; the internet is the largest example of a WAN.

Networks allow shared access to files, centralised data storage, and communication tools (e.g. shared email systems) across an organisation.

7. The internet and email in business

The **internet** is a global network of interconnected computer networks, enabling businesses to communicate, research, advertise, and increasingly, transact online (e-commerce, online banking). **Email** is one of the most widely used business communication tools — fast, provides a written record, but requires care with tone (since non-verbal cues are absent) and with security (phishing attempts often arrive via email).

8. Information and data security

Protecting business data and information systems involves:

- **Passwords and access controls** — restricting system access to authorised users only, ideally with strong, regularly changed passwords.
- **Antivirus/anti-malware software** — protecting against malicious programs that could damage systems or steal data.
- **Backups** — regularly copying data to a separate location/medium, so it can be recovered if the original is lost, corrupted, or affected by a ransomware attack.
- **Firewalls** — controlling and monitoring incoming/outgoing network traffic to block unauthorised access.
- **User awareness** — training staff to recognise phishing attempts and avoid careless practices (e.g. sharing passwords, using unsecured public networks for sensitive work) since human error is a very common cause of security breaches.

9. Full worked question — a simple spreadsheet formula scenario

Scenario. A spreadsheet lists five days of sales in cells B2 to B6: 4,000; 6,500; 3,200; 7,800; 5,000 (all figures in KSh).

Required: (a) Write the formula that would be entered in cell B7 to compute the total sales for the five days. (b) Write the formula that would be entered in cell B8 to compute the average daily sales. (c) State the resulting values for B7 and B8.

Solution:

(a) Formula in B7: `=SUM(B2:B6)`

(b) Formula in B8: `=AVERAGE(B2:B6)`

(c) Total (B7) = 4,000 + 6,500 + 3,200 + 7,800 + 5,000 = **KSh 26,500**

Average (B8) = 26,500 ÷ 5 = **KSh 5,300**

10. Practice examination — KASNEB-style paper (original, not a past paper)

Note on format. This section is laid out the way a KASNEB ATD examination paper is laid out — a timed paper with five compulsory 20-mark questions — purely so you can practise under realistic exam conditions. Every question below is an original scenario written for this document. **It is not a reproduction of any actual KASNEB past examination paper.**

ATD LEVEL I — INFORMATION COMMUNICATION TECHNOLOGY (PRACTICE PAPER)

TIME ALLOWED: 3 HOURS

Instructions to candidates: Answer ALL FIVE questions. Each question carries 20 marks. Show all workings clearly.

QUESTION ONE

(a) Distinguish between computer hardware and software, giving TWO examples of each. (12 marks)

(b) Distinguish between system software and application software. (8 marks)

(Total: 20 marks)

QUESTION TWO

(a) Distinguish between "data" and "information," giving one example of each. (8 marks)

(b) Outline FOUR qualities of good information. (8 marks)

(c) Outline TWO input devices and TWO output devices. (4 marks)

(Total: 20 marks)

QUESTION THREE

A spreadsheet lists four weeks of expenses in cells C2 to C5: 12,000; 15,500; 9,800; 13,200 (KSh).

Required:

(a) Write the formula for cell C6 to compute total expenses. (3 marks)

(b) Write the formula for cell C7 to compute average weekly expenses. (3 marks)

(c) Compute the values that would appear in C6 and C7. (6 marks)

(d) Outline FOUR common application software packages used in business, with one use of each. (8 marks)

(Total: 20 marks)

QUESTION FOUR

(a) Distinguish between a LAN and a WAN. (6 marks)

(b) Outline THREE benefits a business gains from networking its computers. (9 marks)

(c) Outline TWO risks of using email for business communication. (5 marks)

(Total: 20 marks)

QUESTION FIVE

(a) Outline FOUR measures a business can take to protect its information and data security. *(16 marks)*

(b) Explain why user awareness/training is considered an important part of information security. *(4 marks)*

(Total: 20 marks)

11. Marking guide — model answers to the practice examination

QUESTION ONE

(a) *(12 marks)* — Hardware is the physical, tangible components of a computer (e.g. keyboard, monitor); software is the intangible instructions/programs directing hardware (e.g. an operating system, a spreadsheet program) — 3 marks each for definition/example set.

(b) *(8 marks)* — System software manages hardware resources and provides the platform for other software to run (e.g. an operating system); application software is designed for specific user tasks (e.g. a word processor or accounting package).

QUESTION TWO

(a) *(8 marks)* — Data is raw, unprocessed facts with no inherent meaning (e.g. a list of individual transactions); information is data processed and organised to become meaningful for decision-making (e.g. a monthly sales summary).

(b) *(8 marks — 2 marks each)* — Accurate, timely, relevant, complete (understandable/appropriately presented also acceptable).

(c) *(4 marks — 1 mark each)* — Input: keyboard, mouse (or scanner). Output: monitor, printer (or speakers).

QUESTION THREE

(a) *(3 marks)* — `=SUM(C2:C5)`

(b) *(3 marks)* — `=AVERAGE(C2:C5)`

(c) *(6 marks)* — Total = 12,000 + 15,500 + 9,800 + 13,200 = **KSh 50,500**. Average = 50,500 ÷ 4 = **KSh 12,625**.

(d) *(8 marks — 2 marks each)* — Word processor (documents/letters); spreadsheet (budgets/calculations); accounting/ERP package (recording transactions); presentation software (slides for meetings) — database software also acceptable.

QUESTION FOUR

(a) (6 marks) — A LAN covers a small geographic area (e.g. one office building); a WAN covers a large geographic area, potentially connecting multiple cities/countries.

(b) (9 marks — 3 marks each) — Shared access to files; centralised data storage; shared hardware resources (e.g. a shared printer) — shared communication tools also acceptable.

(c) (5 marks) — Risk of phishing/malicious attachments; risk of miscommunication due to absence of non-verbal cues/tone (any two, appropriately explained).

QUESTION FIVE

(a) (16 marks — 4 marks each) — Passwords/access controls; antivirus/anti-malware software; regular backups; firewalls.

(b) (4 marks) — Because human error (e.g. falling for phishing, sharing passwords, careless use of unsecured networks) is a common cause of security breaches, so trained, alert users are a key defence alongside technical controls.

12. Exam technique and revision checklist

- Hardware vs software questions: always give at least one concrete example alongside the definition — a bare definition without an example often earns partial marks only.
- Spreadsheet formula questions: write the exact formula syntax (e.g. `=SUM(B2:B6)`), not just a description of "adding them up" — precise syntax is usually what's specifically being tested.
- Data vs information: remember information is data that has been processed/organised to become meaningful — this distinction is very commonly tested.
- Security questions: name specific measures (passwords, antivirus, backups, firewalls) rather than a vague "make it secure" — precision earns marks.
- LAN vs WAN: the distinguishing factor is geographic coverage — keep the contrast simple and direct.

Quick revision — one line per topic:

1. Hardware = physical components; software = the instructions/programs.
2. System software runs the platform; application software performs specific user tasks.
3. Data = raw unprocessed facts; information = processed, meaningful data.
4. Good information: accurate, timely, relevant, complete, understandable.
5. Spreadsheet basics: `=SUM(range)` totals; `=AVERAGE(range)` computes the mean.
6. LAN = small area network; WAN = large area network (internet is the largest WAN).
7. Email: fast and documented, but watch for phishing and tone/miscommunication risk.

8. Security measures: passwords/access control, antivirus, backups, firewalls, user awareness training.

End of study notes. This document is intended as original exam-preparation material and should be used alongside the current official KASNEB syllabus and examiner's reports for the paper.